FF-B-584E
June 17, 1976
SUPERSEDING
FF-B-584D
March 18, 1970

FEDERAL SPECIFICATION

BOLTS, FINNED NECK; KEY HEAD; MACHINE; RIBBED NECK; SQUARE NECK; TEE HEAD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers bolts, 0.190 inch in diameter and larger, with heads which are either semispherical or designed to set flush with the surface of the item being fastened.

1.2 Classification. Bolts covered by this specification shall be of the following types, classes, styles, and forms as specified (see 6.2):

Type I - Bolt, square neck.

Class 1 - Semispherical head.

Style A - Regular.
Style B - Short neck.
Style C - Step.

Class 2 - Flush head, countersunk.

Style A - Regular.
Style B - Elevator.
Style C - Plow bolt.

Form 1 - Regular head.
Form 2 - Repair head.

Type II - Bolt, machine.

Class 1 - Button head.
Class 2 - Round countersunk, unslotted.
Class 3 - Square countersunk.

Form 1 - Regular head.
Form 2 - Repair head.

FSC 5306

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Type III - Bolt ribbed neck.

Type IV - Bolt, finned neck.

Class 1 - Regular.
Class 2 - Countersunk, unslotted.

Type V - Bolt, tee head.

Type VI - Bolt, key head.

Class 1 - Countersunk, reverse key.

Form 1 - Regular head.
Form 2 - Repair head.

Class 2 - Countersunk, heavy key.

Form 1 - Regular head.
Form 2 - Repair head.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

FF-N-836 - Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
QQ-A-225/6 - Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 2024.
QQ-B-637 - Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products With Finished Edges (Bar, Flat Wire, and Strip).
QQ-B-639 - Brass, Naval: Flat Products (Plate, Bar, Sheet, and Strip).
QQ-B-728 - Bronze Manganese; Rod, Shapes, Forgings, and Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate).
QQ-B-750 - Bronze, Phosphor; Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.
QQ-C-591 - Copper-Silicon, Copper-Zinc-Silicon, and Copper-Nickel-Silicon Alloys: Rod, Wire, Shapes, Forgings, and Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate).

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QQ-N-281 - Nickel-Copper-Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections).
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-763 - Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for.
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 151 - Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Reginal Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-H-6088 - Heat Treatment of Aluminum Alloys.
MIL-P-16232 - Phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connections with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H-28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

American National Standards Institute (ANSI) Standards:

B18.5 - Round Head Bolts.
B18.9 - Plow Bolts.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM) Standards:

B117 - Salt Spray (Fog) Testing.
E8 - Tension Testing of Metallic Materials.
E10 - Brinell Hardness of Metallic Materials.
E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Carbon steel. Unless otherwise specified (see 6.2), the bolts shall be made of carbon or alloy steel conforming to the requirements of Table I.

3.1.2 Corrosion-resistant steel. When specified, bolts of corrosion-resistant steel shall be made of material conforming to class 302B, 304B, 316B, or QQ-S-763.

3.1.3 Brass (Naval Brass). When specified, bolts shall be made of Naval Brass conforming to QQ-B-637 or QQ-B-639. Cold worked brass bolts shall be stress-relieved by heating at 600° Fahrenheit for a minimum of 1 hour per inch of shank diameter.

3.1.4 Aluminum. When specified, bolts shall be made of aluminum alloy conforming to QQ-A-225/6 and heat-treated in accordance with MIL-H-6088.

3.1.5 Nickel-copper. When specified, bolts shall be made of nickel-copper alloy in accordance with QQ-N-281.

3.1.6 Bolts of other materials. When specified, bolts shall be made of alloys covered by QQ-B-728, QQ-B-750, QQ-C-591, QQ-N-286, or as otherwise specified.

3.1.7 Protective coating. Unless otherwise specified (see 6.2), bolts shall be furnished uncoated, except for black oxide coating, which is acceptable. Bolts required to be plated or phosphate-coated shall be to one of the following:

(a) Cadmium plating conforming to QQ-P-416, type II, class 3.
(b) Zinc plating conforming to QQ-Z-325, type II, class 3.
(c) Phosphate coating conforming to MIL-P-16232, type I, class 2.

3.1.7.1 Hydrogen embrittlement. Bolts required to be plated or phosphate-coated shall be subjected to a relief treatment in accordance with the applicable plating or coating specification. This treatment shall follow the plating or coating operation, to reduce to a minimum any tendency to brittleness resulting from these processes.

3.2 Fabrication. Unless otherwise specified (see 6.2), the method of fabrication shall be at the option of the manufacturer.

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TABLE I. Mechanical and chemical properties of carbon and alloy steel bolts

Material	Nominal size (inches)	Proof load stress	Yield* Strength min psi	Tensile Strength min psi	Elongation in 2 inch min percent	Reduction in area percent
Commercial	All sizes	-	-	55,000	14	35
Low Carbon	To 1/2 encl.	55,000	57,000	69,000	14	35
	Over 1/2 to 3/4 incl.	52,000	54,000	64,000	14	35
	Over 3/4	28,000	29,000	55,000	14	35
Medium carbon	To 3/4 incl.	85,000	88,000	120,000	14	35
	Over 3/4 to 1 incl.	78,000	81,000	115,000	14	35
	Over 1 to 1-1/2 incl	74,000	77,000	105,000	14	35
	Over 1-1/2	55,000	55,000	90,000	14	35
Alloy	To 1-1/2 incl.	120,000	124,000	150,000	14	35

		Hardness		Chemical composition		
Material	Nominal size (inches)	Brinell	Rockwell	Carbon (percent)	Phosphorus maximum (percent)	Sulfur maximum (percent)
Commercial	All sizes	207 max.	95B max.	--	0.05	0.06
Low carbon	To 1/2 incl.	241 max.	100B max.	0.28 max.**	0.04	0.05
	Over 1/2 to 3/4 incl.	241 max.	100B max.	0.28 max.**	0.04	0.05
	Over 3/4	207 max.	95B max.	0.55 max.	0.04	0.05
Medium carbon	To 3/4 incl.	241-302	23-32C	0.28-0.55	0.04	0.05
	Over 3/4 to 1 incl.	235-302	22-32C	0.28-0.55	0.04	0.05
	Over 1 to 1-1/2 incl.	223-285	19-30C	0.28-0.55	0.04	0.05
	Over 1-1/2	183-235	--	0.28-0.55	0.04	0.05
Alloy	To 1-1/2 incl.	302-352	32-38C	0.28-0.55	0.04	0.05

* This test is run at 0.2 percent offset.
**For bolts over 6 inches long, carbon may be 0.55 max.

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3.2.1 Body blanks. Body blanks shall be made from wire or rods by hot or cold upsetting, extrusion, or machining at the option of the manufacturer. Cold upset blanks shall be stress-relieved, except when heat treated to meet the specified machanical properties.

3.3 Detail requirements. Bolt lengths, bolt length tolerance, thread lengths, fillets, and head dimensions shall conform to the applicable tables of ANSI B18.5 and B18.9 for the specific type, class, style and form of bolt required. Threads shall conform to NBS H28.

3.4 Nuts. When specified (see 6.2), the bolt shall be furnished with a mating nut that conforms to FF-N-836. Unless otherwise specified (see 6.2), the nut shall conform to type 1, style 1 of FF-N-836.

3.5 Workmanship. The quality of workmanship shall be such as to produce bolts that meet the requirements of this specification and insure proper functioning of all parts of the unit.-

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.2.2).
(b) Preparation for delivery inspection (see 4.6).

4.2.1 Quality conformance inspection. Quality conformance inspection shall be performed on the SAMPLE bolts selected in accordance with 4.4. This inspection shall include the examination of 4.3.1 and the tests of 4.3.2.

4.2.2 Inspection lot. All units of the same size, type, form, coating material, style, thread and class, offered to the Government at one time, shall be considered a lot for purpose of inspection. The sample unit shall be one complete bolt.

4.3 Sampling. A random sample of bolts shall be selected from each lot in accordance with MIL-STD-105.

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4.3.1 Sampling for examination. Examination of the bolts shall be based on inspection level I, and an Acceptable Quality Level (AQL), of 2.5 percent defective for major defects and 4.0 percent defective for minor defects

4.3.2 Sampling for tests. Tests of the bolts shall be based on inspection level S2 and an AQL of 2.5 percent defective.

4.4 Examination. Each sample selected in accordance with 4.3.1 shall be examined for affects listed in Table II.

TABLE II. Classification of defects

Examine	Defects	Major	Minor
End item	Threads not as specified	X	
	Pitch incorrect	X	
	Dimensions of head characteristics incorrect	X	
	Length of bolts incorrect		X
	Thread lengths incorrect		X
	Finish not as specified		X
	Fillets under bolt head incorrect		X
	Damage or defects no affecting function or serviceability		X
	Type, class, style, and form not as specficed		X

4.5 Tests. Each sample selected in accordance with 4.3.2 shall be tested to determine compliance with this specification. Tests shall be conducted as specified in 4.5.1 through 4.5.6.

4.5.1 Yield strength. When determination of yield strength is required the methods used shall be in accordance with ASTM E8.

4.5.2 Hardness. When the length of bolt or thread length is too short for a tension test, a hardness test in accordance with ASTM E10 or E18 shall be substituted.

4.5.3 Chemical analysis. Samples selected in accordance with 4.3 shall be tested for chemical composition, in accordance with method 111.2 of Fed. Test Method Std. No. 151. A certificate of compliance may be accepted in lieu of this test as evidence that the material is as specified herein.

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4.5.4 Salt-spray test. When specified (see 6.2), the salt-spray test described in ASTM B117 shall be applied to samples selected in accordance with 4.3. The test for zinc- or cadmium-plating shall extend for a period of 32 hours.

4.5.5 Tensile strength. Samples selected in accordance with 4.3 shall be tested for tensile strength in accordance with ASTM E8.

4.5.6 Turned shank specimens. To test bolts having tensile properties greater than can be measured on available test equipment, the shank or test specimen shall be turned down around the axis, to gage dimensions specified in ASTM E8. To test bolts or test specimens too large for full size testing, the specimen shall be turned, have its axis midway between the axis of the shank and the surface of the shank, as illustrated in figure 4 of ASTM E8.

4.6 Preparation for delivery inspection. The preservation, packaging, packing, and marking shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The bolts shall be preserved and packaged in accordance with the applicable level A requirements of PPP-H-1581.

5.1.2 Level C. The bolts shall be preserved and packaged in accordance with the contractor's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The bolts shall be packed in accordance with the applicable level A requirements of PPP-H-1581.

5.2.2 Level B. The bolts shall be packed in accordance with the applicable level B requirements of PPP-H-1581.

5.2.3 Level C. The bolts shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or with National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable (see 6.2).

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6. NOTES

6.1 Intended use. This specification covers various types of bolts which are used with nuts as fastening devices.

6.2 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, and form of bolt required (see 1.2).
(c) Material, if different (see 3.1.1).
(d) Type of protective coating, if required (see 3.1.7).
(e) Method of fabrication, if different (see 3.2).
(f) Nut, if required (see 3.4).
(g) Type and style of nut, if different (see 3.4).
(h) Salt-spray test, when required (see 4.5.4).
(i) Levels of packaging and packing required (see 5.1 and 5.2).
(j) Marking required (see 5.3).

MILITARY INTEREST:

Custodians:

Army - WC
Air Force - 82

Review activities:

Army - MI, MU
Navy - SH
Air Force - 82

User activities:

Army - AT, AV
Navy - MC

CIVIL AGENCY INTEREST:

AGR - AFS
GSA - FSS
JUSTICE - FPI

Preparing activity:

Navy - YD

Project No. 5306-0574

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35 cents each.

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